



Build guide for the Quatta AI layer

This page connects the architecture flowcharts to concrete setup work. It defines what to build, which services to use, when to use MCP, RAG, Vertex AI, Document AI and LangGraph, and what a production-ready customer pilot must prove.

[Full overview doc](#)[Governance doc](#)[Roadmap doc](#)

MCP

Use for standardized agent tools with schemas, permission checks and audit logs. Keep connector logic reusable outside MCP.

RAG

Use when answers depend on contracts, SOPs, claims evidence, PDFs, scans or policies. Structured ERP/WMS data should stay structured.

Vertex AI

Use Gemini as the default enterprise model provider on GCP. Keep an OpenAI/Azure OpenAI provider adapter for customer preference.

Human gates

Agents recommend and draft. Humans approve claim sending, operational corrections and high-risk actions.

IMPLEMENTATION SEQUENCE

Build the platform in this order

1

Platform skeleton

FastAPI on Cloud Run, PostgreSQL, health endpoints, Entra token validation and OpenTelemetry.

[Foundation](#)**2**

Read-only connectors

ERP, WMS, SQL, files, SharePoint, APIs and documents through normalized connector

Data access

3

Retrieval and RAG

Structured SQL/API retrieval plus Document AI OCR, chunking, embeddings, pgvector and Context Builder.

Context

4

AI gateway

Model routing, prompt policy, data minimization, safety checks, validation and provider abstraction.

Control plane

5

Inventory Agent MVP

LangGraph workflow for stock variance analysis, reconciliation, confidence, citations and recommended actions.

Use case 1

6

Claims Agent MVP

Eligibility, liability, evidence collection, draft claim email, human review and status tracking.

Use case 2

STACK DECISIONS

Services and when to use them

SERVICE	USE IT FOR	CRITICAL SETUP DETAIL
FastAPI	Quatta API, AI Gateway, connector endpoints and agent endpoints.	Use typed request/response models and structured error contracts.
Cloud Run	Stateless backend hosting and worker-style ingestion services.	Separate runtime service accounts by API, ingestion and export workloads.

SERVICE	USE IT FOR	CRITICAL SETUP DETAIL
Microsoft Entra ID	User authentication and enterprise group/app-role mapping.	Map Entra identity to Quatta tenant, role, site, customer and item permissions.
PostgreSQL	Audit, config, normalized metadata, cache, human review state.	Enforce tenant filters in every table and query path.
pgvector	Vector index for SOPs, contracts, evidence and document chunks.	Store tenant, ACL, document version and source timestamp with every chunk.
Vertex AI Gemini	Primary model inference for grounded answers and agent reasoning.	Route all model calls through the AI Gateway, never directly from tools.
Document AI	OCR for scans, PDFs, contracts, PODs and claims evidence.	Preserve document ID, version, ACL and source timestamp through indexing.
LangGraph	Multi-step workflows with branching, retries and human checkpoints.	Persist state and audit every node execution.
MCP	Agent-callable connector tools with discoverable schemas.	Use read-only tools by default and enforce tool policy per tenant and role.
OpenTelemetry	Trace requests, connectors, retrieval, model calls and human review.	Track latency, errors, connector health, retrieval quality and model cost.

CLICKABLE FLOWCHART COMPANION

Implementation docs per flowchart

01 High-Level Platform

Cloud Run, FastAPI, Entra ID, AI Gateway, PostgreSQL audit, OpenTelemetry and tenant isolation.

02 Connector Layer

Read-only adapters, source credentials, caching, normalized output, connector health and MCP tool shape.

03 Retrieval + RAG

Structured retrieval, Document AI, chunking, embeddings, pgvector, metadata filters and context building.

04 Inventory Agent

Stock variance workflow, ERP/WMS reconciliation, pattern analysis, financial impact and recommendations.

05 Claims Agent

Claim eligibility, liability calculation, evidence summary, draft email, human review and approved send.

PILOT READINESS

What must be true before a customer pilot

Access control

- Entra sign-in works.
- Tenant and role filters pass tests.
- No cross-tenant vector retrieval.

Source confidence

- At least one structured source connected.
- At least one document source indexed.
- Freshness warnings are visible.

Agent quality

- Known cases pass evaluation.
- Low confidence escalates.
- Answers include citations.

Operations

- Audit reconstructs every answer.
- Dashboards and alerts exist.
- Runbooks are ready.